DESeq2 Median of Ratios

What is "library size"?

When you sequence RNA, each sample gets read out into millions of short sequence fragments ("reads"). Each read gets mapped to a gene, and you count how many reads landed on each gene. **Library size** = the total number of reads for that sample (sum of all gene counts in that column).

So if Sample A has 20 million total reads and Sample B has 30 million total reads, Sample B just has a bigger "library" — not necessarily because its genes are more active, but maybe because the machine sequenced it more deeply, or the RNA extraction was more efficient, or a hundred other technical reasons.

The problem: if Gene X has 100 counts in Sample A and 150 counts in Sample B, is that gene really more expressed in B? Or is B's library just 1.5x bigger overall, so *everything* looks 1.5x bigger? You can't tell until you correct for library size. That correction is what "normalization" means here.

Why not just divide by total library size (like CPM)?

The naive fix: divide every gene's count by the sample's total read count (this is what CPM/TPM do). Sounds reasonable — except:

Imagine one sample has 3 genes that are cancer-related and massively overexpressed — like 40% of all reads in that sample come from just those 3 genes. Now the *total* read count for that sample is inflated by those 3 genes alone. If you divide everything by that inflated total, you artificially shrink every other gene's normalized value, even though those other genes didn't actually change. A few extreme genes distort the correction for every gene.

DESeq2's method is built to avoid exactly this trap.

Median-of-ratios, step by step with real numbers

Say you have 3 genes and 3 samples (raw counts):

Gene	Sample A	Sample B	Sample C
G1	10	20	30
G2	100	200	300
G3	1000	500	1500

Step 1 — build a "pseudo-reference sample" For each gene, take the geometric mean across all samples (geometric mean = multiply the values together, then take the nth root — it's like an average that's less sensitive to one huge outlier than a normal average).

- G1 reference = geometric mean(10, 20, 30) ≈ 18.2
- G2 reference = geometric mean(100, 200, 300) ≈ 182
- G3 reference = geometric mean(1000, 500, 1500) ≈ 953

This "pseudo-reference" isn't a real sample — it's an imaginary "typical" sample built gene-by-gene, used purely as a yardstick to compare every real sample against.

Step 2 — for each real sample, divide each gene's count by the reference value for that gene

For Sample A:

- G1: 10 / 18.2 = 0.55
- G2: 100 / 182 = 0.55
- G3: 1000 / 953 = 1.05

Step 3 — take the median of those ratios for that sample Median of (0.55, 0.55, 1.05) = **0.55**. That's Sample A's size factor.

Do the same for Sample B and Sample C, and you get one size factor per sample.

Step 4 — divide each sample's raw counts by its size factor That gives you normalized counts, on a comparable scale across samples.

Why the *median* (not mean) of the ratios?

This is the clever part. Suppose one gene (like G3 above, or a cancer gene) has a huge, real biological difference between samples — its ratio for that sample will be an outlier (very high or very low) compared to the ratios of the thousands of other genes, which mostly don't change.

- If you averaged all the ratios, that one extreme gene would drag the whole sample's size factor off course.
- The **median** ignores extreme values — it just picks the middle ratio. Since in a real dataset the vast majority of genes are *not* differentially expressed, the median ratio reflects the "typical" gene's behavior, which is exactly what you want your normalization factor to represent.

So: **library size** = raw problem (samples sequenced to different depths). **Median-of-ratios** = DESeq2's clever fix that's robust to a few genes being wildly different, by trusting the *median* gene rather than the *total* or the *mean*. **Pseudo-reference sample** = just a made-up yardstick, one number per gene, used only to compute those ratios.